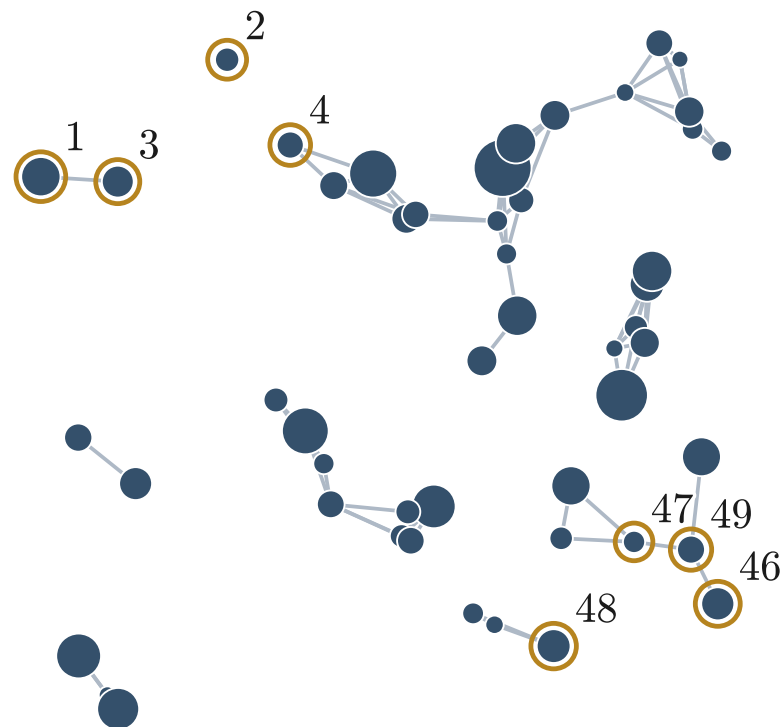


Invariance and equivariance under permutation

Initial ordering



$$\Omega_{M,0} = 0.31$$

$$\vec{M}_{\star,0} = [11.2, 10.1, \dots, 10.4, 10.5]$$

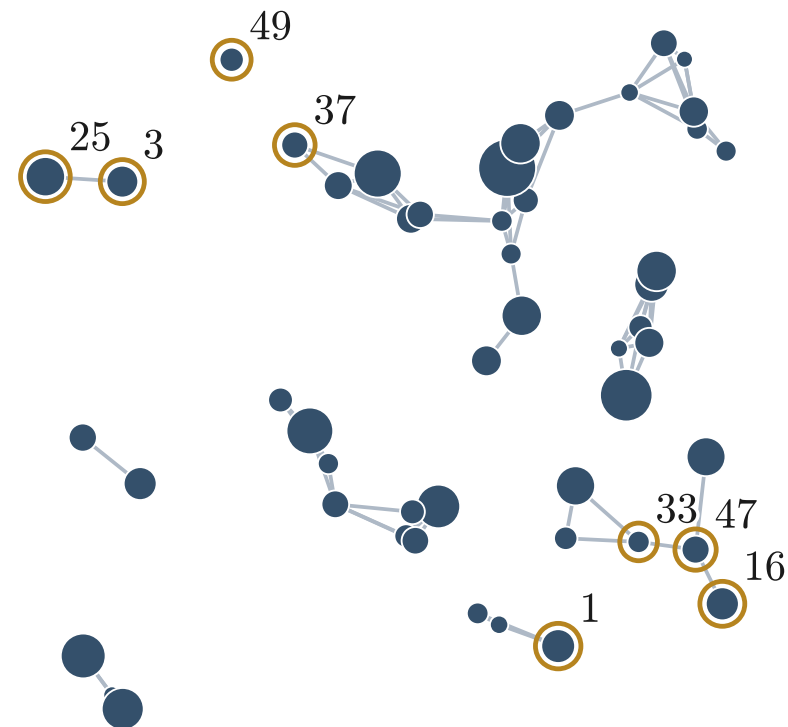
P

Permutation simply
reindexes nodes

Permutation simply reindexes nodes		Permuted index				
		1	...	47	48	49
Initial index	1	⋮	⋮	⋮	⋮	⋮
	2	0	...	0	0	1
	⋮	⋮	⋮	⋮	⋮	⋮
	48	1	...	0	0	0
	49	0	...	1	0	0

A 1 at ij permutes
node i to j

Permuted (reordered)



$$P(\Omega_{M,0}) = \Omega_{M,0} = 0.31$$

$$\vec{M}_{\star,P} = P(\vec{M}_{\star,0}) = [10.4, \dots, 11.2, \dots, 10.5, \dots, 10.1]$$

Permute nodes with P : Global Ω_M is **invariant** ($g(P\mathbf{x})=g(\mathbf{x})$); the per-halo $M_{\star}$ are **equivariant** ($f(P\mathbf{x})=Pf(\mathbf{x})$).